

Secondary Examination, 2016

Mathematics

Time- $3\frac{1}{4}$ Hours

M.M- 70

GENERAL INSTRUCTIONS TO THE EXAMINEES:

- 1) There are in all seven questions in this question paper.
- 2) All the questions are compulsory.
- 3) In the beginning of each, the number of parts to be attempted has been clearly mentioned.
- 4) Marks allotted to the questions are indicated against them.
- 5) Start from the first question and proceed to the last one.
- 6) Do not waste your time over a question you cannot solve

1. a) If $x = 2 - \sqrt{3}$, find the value of $x^2 + \frac{1}{x^2}$.

Sol.

Simplify the expression,

$$x = 2 - \sqrt{3}$$

Similarly

$$\frac{1}{x} = 2 + \sqrt{3}$$

Therefore,

$$x + \frac{1}{x} = 2 - \sqrt{3} + 2 + \sqrt{3}$$

$$x + \frac{1}{x} = 4$$

Squaring both side,

$$\left(x + \frac{1}{x}\right)^2 = 4^2$$

$$x^2 + \frac{1}{x^2} + 2x \cdot \frac{1}{x} = 16$$

$$x^2 + \frac{1}{x^2} + 2 = 16$$

$$x^2 + \frac{1}{x^2} = 16 - 2$$

$$x^2 + \frac{1}{x^2} = 14$$

Hence, the value of 14.

b) If $\sec \theta = 2$, find the value of θ .

i) $\frac{\pi}{2}$ (ii) $\frac{\pi}{3}$ (iii) $\frac{\pi}{4}$ (iv) $\frac{\pi}{6}$

sol.

Simplify the expression,

We know that,

$$\sec \theta = 2$$

$$\sec \theta = 60^\circ$$

Therefore,

$$\theta = 60^\circ$$

$$\theta = \frac{60^\circ}{180^\circ} \cdot \pi$$

$$\theta = \frac{\pi}{3}$$

Hence, the value of option (ii) is $\frac{\pi}{3}$.

c) If $a+b+c=0$, find the value of $\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab}$.

Sol.

Simplify the expression,

$$\frac{a^2}{bc} + \frac{b^2}{ca} + \frac{c^2}{ab} = \frac{a^3 + b^3 + c^3}{abc}$$

We know that,

$$a^3 + b^3 + c^3 - 3abc = 0$$

$$a^3 + b^3 + c^3 = 3abc$$

So,

$$\frac{3abc}{abc} = 3$$

Hence, the value of 3.

c) find the 1 to 9 nature number of Arithmetic progress.

Sol.

Simplify the expression,

1, 2, 3, 4, 5, 6, 7, 8, 9 are nature number,

$$A.P = \frac{1+2+3+4+5+6+7+8+9}{9}$$

$$= \frac{45}{9}$$

$$= 5$$

Hence, the value of 5.

d) What is the language of withdrawing bank fund from the account?.

(i) Debit (ii) Credit (iii) saving (iv) none of these.

Sol.

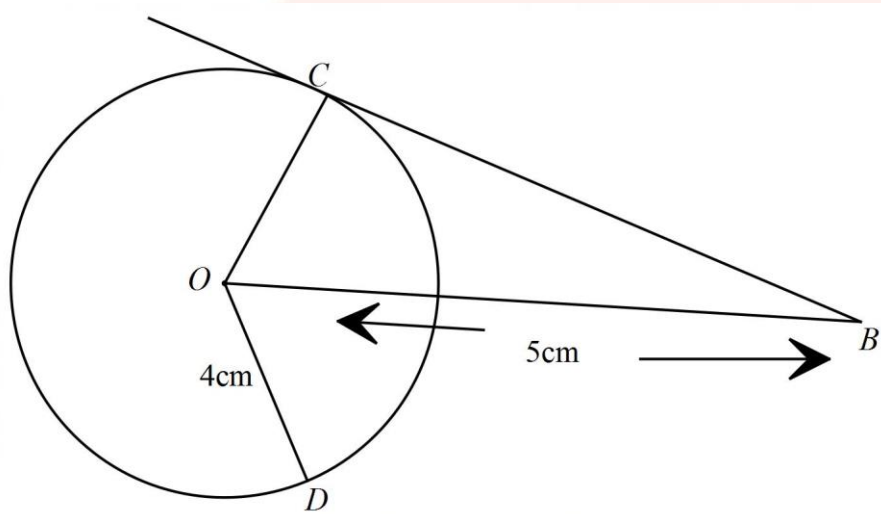
Simplify the expression,

The language of withdrawing fund from the account is called none of these.

Hence, the option (iv)

e) Figure in, $OD=4\text{cm}$ is the radius of centre of O . If $OB=5\text{ cm}$, So find the value of tangent of BC .

(i) 3cm (ii) 4cm (iii) 2cm (iv) 3.5



Sol.

Simplify the expression,

$$OD = OC$$

we get $OD=4\text{cm}$

then,

$$OC=4\text{cm}$$

Therefore,

$$OB^2 = OD^2 + BC^2$$

$$5^2 = 4^2 + BC^2$$

$$BC^2 = 5^2 - 4^2$$

$$BC^2 = 25 - 16$$

$$BC^2 = 9$$

$$BC = 3$$

Hence, the value of BC of 3cm.

2.a) If $A = x + \frac{1}{x}$, find the value of $\frac{1}{A}$.

Sol.

Simplify the expression,

$$\begin{aligned} A &= x + \frac{1}{x} \\ &= \frac{x^2 + 1}{x} \end{aligned}$$

Then,

$$\frac{1}{A} = \frac{x}{x^2 + 1}$$

Hence, the value of $\frac{1}{A} = \frac{x}{x^2 + 1}$.

b) If $A + B = 45^\circ$, So find the value of $\sin A \cos B + \cos A \sin B$.

Sol.

Simplify the expression,

We know that,

$\sin A \cos B + \cos A \sin B$ this the formula of $\sin(A + B)$

Then,

$$\begin{aligned} &\sin 45^\circ \\ &= \frac{1}{\sqrt{2}} \end{aligned}$$

Hence, the value of $\frac{1}{\sqrt{2}}$.

c) Given the number of 6,3,2,6,5,6 and 8 . find the mode.

Sol.

Simplify the expression,

We know that,

More than frequency of number is called mode.

Then,

The number given are high frequency of 6 ,

Hence, the value of 6.

d) Udit buying a printer in 35,00 whose sell on 12% profit , then find the selling price of printer.

Sol.

Simplify the expression,

We given the cost price of printer is 35,00

$$12\% = \frac{12}{100}$$

$$3500 \times \frac{12}{100} = 35 \times 12 = 420$$

when,

$$3500 + 420 = \text{Rs}3920$$

We get the selling price of printer is Rs=3920

Hence, the value of Rs3920.

3.a) A line drawn through the end of a radius and perpendicular to it is a tangent to the circle.

Sol.

Given : A circle with centre O in which OP is radius and AB is a line through P such that $OP \perp AB$.

To prove : AB is a tangent to the circle at the point P.

Construction : Take a point Q, different from P, on AB. Join OQ.

Proof : we know that the perpendicular distance from a point to a line is the shortest distance between them.

$$\therefore OP \perp AB$$

$$\therefore OP < OQ$$

$\therefore$ Q lies outside the circle [OP is the radius and $OP < OQ$]

Thus, every point on AB, other than P, lies outside the circle.

AB meets the circle at the point P only.

Hence, AB is the tangent to the circle at the point P.

b) Prove that $\frac{\sin 36^\circ}{\cos 54^\circ} + \frac{\cos 36^\circ}{\sin 54^\circ} = 2$

sol.

Prove the expression,

Taking L.H.S

$$\begin{aligned} & \frac{\sin 36^\circ}{\cos 54^\circ} + \frac{\cos 36^\circ}{\sin 54^\circ} \\ & \frac{\sin(90^\circ - 54^\circ)}{\cos 54^\circ} + \frac{\cos(90^\circ - 54^\circ)}{\sin 54^\circ} \end{aligned}$$

We know that,

$$\sin(90^\circ - \theta) = \cos \theta$$

$$\cos(90^\circ - \theta) = \sin \theta$$

Then,

$$\begin{aligned} & = \frac{\sin(90^\circ - 54^\circ)}{\cos 54^\circ} + \frac{\cos(90^\circ - 54^\circ)}{\sin 54^\circ} \\ & = \frac{\cos 54^\circ}{\cos 54^\circ} + \frac{\sin 54^\circ}{\sin 54^\circ} \\ & = 1 + 1 \\ & = 2 \end{aligned}$$

Hence, the proved it.

c) Find the curved surface area of cylinder. Whose the base of radius and height of cylinder are 4cm and 14cm.

sol.

Simplify the expression,

We know that , the cylinder of curved surface area is $2\pi rh$.

$$r = 4\text{cm}$$

$$h = 14\text{cm}$$

$$= 2\pi rh$$

$$= 2\pi \times 4 \times 14$$

$$= 112\pi$$

Hence, the curved surface area of cylinder is 112π .

d) Find the equation of that line , which makes an angle of 60 degree by the y-axis of the intersecting angle -4 and x-axis.

sol.

Simplify the expression,

$$y - \text{axis on} = -4$$

$$\theta = 60^\circ$$

We know that,

$$(y - y_1) = m(x - x_1)$$

$$y - \text{axis} = -4$$

$$x - \text{axis} = 0$$

$$(x_1, y_1) = (0, -4)$$

$$m = \tan 60^\circ$$

$$m = \sqrt{3}$$

$$y - (-4) = \sqrt{3}(x - 0)$$

$$y + 4 = \sqrt{3}x$$

$$\sqrt{3}x - y = 4$$

Hence, the value of 4.

4.a) Find the volume of right circular cone. Whose height of 3.0cm and radius of base is 4.0cm.

Sol.

Simplify the expression,

We know that the volume of cone is $v = \frac{1}{3}\pi r^2 h$.

$$h = 3.0\text{cm}$$

$$r = 4.0\text{cm}$$

$$= \frac{1}{3}\pi \times 4^2 \times 3$$

$$= 64\pi$$

Hence, the value of volume of cone 64π .

b) If $\tan A = \frac{1}{\sqrt{3}}$, $\tan B = \sqrt{3}$. Then find the value of $A+B$.

sol.

Simplify the expression,

$$\tan A = \frac{1}{\sqrt{3}} = \tan 30^\circ$$

$$\tan B = \sqrt{3} = \tan 60^\circ$$

$$\tan A = 30^\circ$$

$$\tan B = 60^\circ$$

then,

$$\tan(A + B) = 30^\circ + 60^\circ$$

$$\tan(A + B) = 90^\circ$$

$$\tan(A + B) = 1$$

$$A + B = 1$$

Hence, the value of $A+B=1$.

c) If $x = \frac{a}{a+b}$ and $y = \frac{b}{a-b}$, then prove that $\frac{1}{x} + \frac{1}{y} = \frac{a^2 + b^2}{ab}$.

Sol.

Prove the expression,

$$x = \frac{a}{a+b}$$

$$\frac{1}{x} = \frac{a+b}{a}$$

$$y = \frac{b}{a-b}$$

$$\frac{1}{y} = \frac{a-b}{b}$$

Simplify further,

$$\frac{1}{x} + \frac{1}{y} = \frac{a+b}{a} + \frac{a-b}{b}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{b(a+b) + a(a-b)}{ab}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{ab + b^2 + a^2 - ab}{ab}$$

$$\frac{1}{x} + \frac{1}{y} = \frac{a^2 + b^2}{ab}$$

Hence, proved it.

d) find the gradient then passing through the two points of (-2, 5) and (6, 4).

Sol.

Simplify the expression,

We know that,

$$M = \frac{y_2 - y_1}{x_2 - x_1}$$

Simplify the further,

(-2, 5), (6, 4)

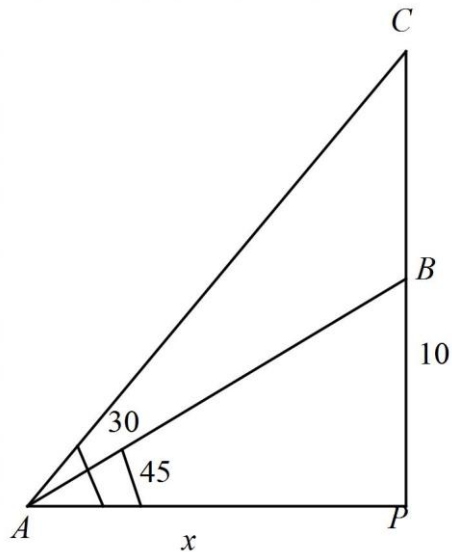
$$M = \frac{4 - 5}{6 - (-2)}$$

$$M = \frac{-1}{8}$$

Hence, the value of gradient is $M = \frac{-1}{8}$.

5. a) An observer standing 10m away from a building notices that the angle of elevation of the top and bottom of a flagstaff on the building are 60° and 45° respectively. Then the height of flagstaff is.

Sol



Simplify the expression,

$\triangle APM$

$$\tan 30^\circ = \frac{10}{x}$$

$$\frac{1}{\sqrt{3}} = \frac{10}{x}$$

$$x = 10\sqrt{3}$$

$\triangle APC$

$$\tan 45^\circ = \frac{10+h}{10\sqrt{3}}$$

$$1 = \frac{10+h}{10\sqrt{3}}$$

$$10\sqrt{3} = 10 + h$$

$$h = 10(\sqrt{3} - 1)$$

$$h = 10(1.732 - 1)$$

$$h = 10(.732)$$

$$h = 7.32m$$

Hence, the flag height of 7.32cm.

b) Find the mean age from the following frequency distribution.

Class-interval	0-10	10-20	20-30	30-40	40-50	50-60
Frequency	7	14	28	26	16	9

Sol.

Simplify the value ,

Interval	Frequency (fi)	Mid-point (xi)	Fixi
0-10	7	5	35
10-20	14	15	210
20-30	28	25	700
30-40	26	35	910
40-50	16	45	720
50-60	9	55	495

$$\sum f_i x_i = 3070$$

$$f_1 = 100$$

$$\text{Mean} = \frac{\sum f_i x_i}{\sum f_i} = \frac{3070}{100} = 30.70$$

Hence, the value of mean is 30.70.

c) Prove that $p^2 = \frac{1+2ab}{a^2+b^2}$.if a line $ax+by+a+b=0$ on root point of perpendicular value of p.

sol.

Simplify the expression,

$$ax+by+a+b=0$$

$$\text{Perpendicular of P} = (p) = \frac{c}{\sqrt{a^2+b^2}}$$

$$p = \frac{a+b}{\sqrt{a^2+b^2}}$$

Squaring both side,

$$p^2 = \frac{a+b}{\sqrt{a^2+b^2}}$$

$$p^2 = \left(\frac{a+b}{\sqrt{a^2+b^2}} \right)^2$$

$$p = \frac{a^2+b^2+2ab}{a^2+b^2}$$

$$p = \frac{a^2+b^2}{a^2+b^2} + \frac{2ab}{a^2+b^2}$$

$$p = 1 + \frac{2ab}{a^2+b^2}$$

Hence, the proved it.

d) (d) Kuldeep In year 2014-15, the annual income of x is 58,0,000 (excluding house rent). He deposits its 9,000 per month in his usual future fund account. He deposits 60,000 in his PDF account. He deposits 80,000 in his PDF account. Kuldeep calculate the income tax payable by X while the maximum limit of savings of saving is 1,50,000. Income tax rates are as follows:

Apart from this 3% education levy is levied on the income tax payable.

	INCOME	INCOMETAX
(i)	UP to Rs 2,00,000	0%
(ii)	Rs 2,00,001 to Rs5,00,000	Rs 2,00,000 increase to income of 10%

Sol.

X of yearly income = Rs 5,580,000 .

Future saving income = per monthly Rs $9,000 \times 12$
Rs 108,000

PPF saving account income = Rs $60,000 + 108,000 + 1,50,000 = \text{Rs } 2,28,000$.

Total saving income = Rs $5,80,000 - 2,28,000 = \text{Rs } 3,52,000$

Calculate the income tax

$$3,52,000 \times 10\%$$

$$= 3,52,000 \times \frac{10}{100} = 52,000$$

Total income tax = 52,000

$$= 52,000 \times 3\%$$

$$= 52,000 \times \frac{3}{100}$$

$$= 1560$$

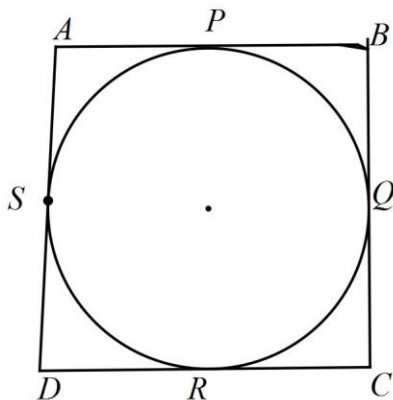
Total pay income tax = $52,000 + 1560 = 53,560$

Hence, the value of total pay income tax 53,560.

6.a) In the given figure, a quad . ABCD drawn to circumscribe a given circle , as shown.

$$AB + CD = BC + DA.$$

Sol.



Prove the expression,

$$AS = AP \dots\dots\dots(1) \quad [\text{tangent of circle}]$$

$$PB = AP \dots\dots\dots(2)$$

$$QC = RC \dots\dots\dots(3)$$

$$SD = RD \dots\dots\dots(4)$$

$$AB + CD$$

$$AP + PB + CR + RD$$

$$AS + BQ + QC + SD$$

$$AS + SD + BQ + QC$$

$$AD + BC$$

Hence, proved it.

b) Find the height of cone. whose curved surface area is $188\frac{1}{7}m^2$ and base of diameter $12m$.

sol.

Simplify the height of cone.

We know that,

$$\pi r l = 188 \frac{4}{7}$$

$$r = 6$$

$$\pi r l = \frac{1320}{7}$$

$$\frac{22}{7} \cdot 6 \cdot l = \frac{1320}{7}$$

$$l = 10$$

$$l^2 = r^2 + h^2$$

$$10^2 = 6^2 + h^2$$

$$h^2 = 10^2 - 6^2$$

$$h^2 = 100 - 36$$

$$h^2 = 64$$

$$h = 8$$

Hence, the value of cone of height is 8.

c) Find the height of the intersect at the axis by line $x \cos \alpha + y \sin \alpha = \sin 2\alpha$.

sol.

Simplify the expression,

$$x \cos \alpha + y \sin \alpha = \sin 2\alpha$$

Divided by both side $\sin 2\alpha$.

$$\frac{x \cos \alpha}{\sin 2\alpha} + \frac{y \sin \alpha}{\sin 2\alpha} = \frac{\sin 2\alpha}{\sin 2\alpha}$$

$$\frac{x \cos \alpha}{2 \sin \alpha \cos \alpha} + \frac{y \sin \alpha}{2 \sin \alpha \cos \alpha} = 1$$

$$x - axix = 2 \sin \alpha$$

$$y - axix = 2 \cos \alpha$$

d) Prove that

$$(\cos A + \cos B)^2 + (\sin A - \sin B)^2 = 4 \cos^2 \frac{A+B}{2}.$$

Prove the expression,

Taking L.H.S

$$\begin{aligned} & (\cos A + \cos B)^2 + (\sin A - \sin B)^2 = \cos^2 A + \cos^2 B + 2 \cos A \cos B + \sin^2 A + \sin^2 B - 2 \sin A \sin B \\ & (\cos^2 A + \sin^2 A) + (\cos^2 B + \sin^2 B) + 2 \cos A \cos B - 2 \sin A \sin B \\ & 2 + \cos(A+B) + \cos(A-B) - [\cos(A-B) - \cos(A+B)] \\ & 2 + \cos(A+B) + \cos(A-B) - \cos(A-B) + \cos(A+B) \\ & 2 + 2(\cos(A+B)) \\ & 2(1 + \cos(A+B)) \\ & 2 \left[1 + 2 \cos^2 \frac{(A+B)}{2} - 1 \right] \\ & 4 \cos^2 \frac{(A+B)}{2} \end{aligned}$$

Hence, proved it.

7.a) Find the numbers .The two positive even numbers square of sum is 244.

Sol.

Simplify the expression,

Let two even number are (x),(x+2).

$$\begin{aligned} & (x^2) + (x+2)^2 = 244 \\ & x^2 + x^2 + 4 + 4x = 240 \\ & 2x^2 + 4x - 240 = 0 \\ & 2(x^2 + 2x - 120) = 0 \\ & x^2 + 2x - 120 = 0 \\ & x^2 + 12x - 10x - 120 = 0 \\ & x(x+12) - 10(x+12) = 0 \\ & (x-10)(x+12) = 0 \\ & x-10 = 0 \\ & x = 10 \\ & x+12 = 0 \\ & x = -12 \end{aligned}$$

Putting the value of x .

$$\begin{aligned}x &= 10 \\&= x + 2 \\&= 10 + 2 \\&= 12\end{aligned}$$

Hence, the number of value are 10 and 12.

OR

Evaluate the value of $\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0$.

Sol.

Simplify the expression,

$$\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 = 0$$

Let,

Adding and subtracting 2 above this equation

$$\left(x^2 + \frac{1}{x^2}\right) - 3\left(x - \frac{1}{x}\right) - 2 + 2 - 2 = 0$$

$$\left(x - \frac{1}{x}\right)^2 - 3\left(x - \frac{1}{x}\right) = 0$$

Let,

$$\left(x - \frac{1}{x}\right) = y$$

$$y^2 - 3y = 0$$

$$y^2 = 3y$$

$$y = 3$$

Hence, the value of 3.

b) A construction quadrilateral PQRS. Whose PQ=5cm, QR=6cm, PR=3.5cm and RS=5cm.

sol.

Below in figure of quadrilateral.

